INCH-POUND
MIL-DTL-32263(AR)
19 OCTOBER 2007
SUPERSEDING
QAT-1451 (AR)
1 MAY 1978

DETAIL SPECIFICATION

COVER, FLEXIBLE ROTATING BAND

Inactive for new design after 15 August 2007

This specification is approved for use by the U.S. Army Armament Research Development and Engineering Center (ARDEC), and is available for use by all Departments and Agencies of the Department of Defense.

1.0 SCOPE

1.1 Scope. This specification covers requirements for the assembly of flexible rotating band cover (FRBC)(see 6.1).

Comments, suggestions, or questions on this document should be addressed to the Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, NJ 07806-5000 or emailed to ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 1398

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2.0 APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to assure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARD

FED-STD-191 - Textile Test Methods

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

(Unless otherwise indicated, Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

12961015 - Webbing
12961016 - Cover, Rotating Band, Flexible

(Copies of these drawings may be requested online at Drawing-Request@pica.army.mil or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

American Society for Testing Materials International (ASTM)

ASTM D3951 – Standard Practice for Commercial Packaging

(Copies of ASTM standards may be ordered online at http://www.astm.org/ or from the ASTM International, 100 Barr Harbor Drive, Conshohocken, PA 19428-2959)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3.0 REQUIREMENTS

3.1 First article inspection. When specified (see 6.2) a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Components and assemblies.

3.2.1 Webbing. The materials, components and assemblies shall be in accordance with dwg. 12961015.

3.2.2 Cover, Rotating Band, Flexible (opened). The materials, components and assemblies shall be in accordance with dwg. 12961016.

3.2.3 Cover, Rotating Band, Flexible (after closing). The materials, components and assemblies shall be in accordance with dwg. 12961016.

3.3 Stretch of rubber (SPANDEX) yarn covered with spun polyester. The rubber covered with spun polyester thread shall be able to be stretched 225 +/- 25 percent of its original length.

3.4 Workmanship. The FRBC shall be uniformly constructed, free from holes, tears, cuts or other defects which might impair the usefulness of the cover for the intended purpose. The cover shall not have grit, dirt or fragments of foreign material.

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4.0 VERIFICATION

TABLE I. Requirement/verification cross reference matrix.

Methods of Verification N/A - Not Applicable 1 - Analysis 2 - Demonstration 3 - Examination 4 - Test							Classes of Verification A- First Article B- Acceptance		
Section 3 Requirements	Description	Verification Method					Verification Class		Section 4 Verification
		N/A	1	2	3	4	A	B	
3.1	First Article inspection				X	X	X		4.2
3.2	Components and assemblies								
3.2.1	Webbing				X	X	X	X	4.3.2.1/4.4.3
3.2.2	Cover, Rotating Band, Flexible (opened)				X		X	X	4.3.2.2
3.2.3	Cover, Rotating Band, Flexible (after closing)				X		X	X	4.3.2.3
3.3	Stretch of rubber yarn covered with spun polyester					X		X	4.4.1/4.4.2
3.4	Workmanship				X		X	X	4.3.2.1 4.3.2.2 4.3.2.3

4.1 Classification of verification. The verification requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection.

4.2.1 First article quantity. First article verification shall be performed on the quantity of items as indicated in table II.

4.2.2 Inspections to be performed. The first article verification shall be performed in accordance with Table II.

4.2.3 First article rejection. If any item fails to comply with requirements, the first article sample shall be rejected.

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TABLE II. First article inspection

EXAMINATION OR TEST	NUMBER OF SAMPLES	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Spun polyester covered spandex rubber Dwg. 12961015 Assembly and stretch of spun polyester covered rubber (Note 1).	8 (Note 2)	3.2.2/3.2.3 /3.3	4.4.1, 4.4.2
Webbing Dwg. 12961015 Examination for defects (Note 1).	8 (Note 2)	3.2.1	4.3.2.1/4.4.3
Flexible rotating band cover Dwg. 12961016 Examination for defects.	10	3.2.2/3.2.3 /3.4	4.3.2.2 and 4.3.2.3

Note 1: Certification acceptable.
Note 2: For purposes of sampling, the unit shall be one roll.

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4.3 Conformance inspection.

4.3.1 Lot formation. Lot formation shall be in accordance with lot formation requirement of MIL-STD-1916.

4.3.2 Classification of characteristics. Conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. When cited herein, attributes sampling inspection shall be conducted in accordance with MIL-STD-1916, using the inspection levels cited in the Classification of Characteristic paragraphs. Acceptance criteria shall be in accordance with MIL-STD-1916. Alternative conformance inspections may be submitted and approved in accordance with MIL-STD-1916.

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.1	**TITLE** Webbing		**SHEET** 1 **OF** 4	**DRAWING NUMBER** 12961015
				NEXT HIGHER ASSEMBLY 12961016
CLASSITIFICAON	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical	None defined			
Major				
101	Material certification. Note 1	100%	3.2.1	Certification
102	Water repellency certification. Note 1	100%	3.2.1	Certification
103	Full width warp ends inspection. Note 2	Level IV	3.2.1	4.4.3
104	Width of webbing.	Level IV	3.2.1	Visual/AIE
105	Abrasion marks resulting in rupture of yarns or in nap sufficient to obscure the identity of any yarn exceeding 10 percent of width or 1 inch in length.	Level IV	3.2.1	Visual/AIE
106	Two or more broken or missing ends regardless of length or a single end exceeding 6 inches in length.	Level IV	3.2.1	Visual/AIE
107	More than one pick broken or missing.	Level IV	3.2.1	Visual
108	Filling bar, course or light, resulting in visible difference in stiffness or thickness of webbing and extending for more than 1/4 inch in the length direction.	Level IV	3.2.1	Visual/AIE

Note 1: A certificate of material conformance shall be submitted by the contractor for each lot utilized. The submission of the documentation shall be specified in the contract.
Note 2: For webbing characteristics listed as Visual or Visual/Measure the defect shall be clearly visible at normal inspection distance (approximately 3 feet). Certification of acceptance from source is acceptable.

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.1	TITLE Webbing		SHEET 2 OF 4	DRAWING NUMBER 12961015
				NEXT HIGHER ASSEMBLY 12961016
CLASSITIFICAON	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major				
109	Cut, hole or tear.	Level IV	3.2.1	Visual
110	Drop ply clearly visible on more than two ends within same length and extending over 9 linear inches or more.	Level IV	3.2.1	Visual/AIE
111	Edges frayed, slack, or otherwise poorly constructed and exceeding 1/4 inch in length.	Level IV	3.2.1	Visual/AIE
112	Three or more floats or skips, 1/2 inch or more in combined warp and filling directions; or single float or skip greater than one inch.	Level IV	3.2.1	Visual/AIE
113	More than three kinks in any linear inch.	Level IV	3.2.1	Visual/AIE
114	Two or more mispicks or double picks across the full length.	Level IV	3.2.1	Visual
115	Two or more slack ends in the same length, jerked in between picks or forming clearly visible loops on the surface.	Level IV	3.2.1	Visual
116	Evidence of smash.	Level IV	3.2.1	Visual
117	Tight end clearly visible up to twelve (12) inches in length.	Level IV	3.2.1	Visual/AIE
118	Wrong draw extending for more than nine (9) inches.	Level IV	3.2.1	Visual/AIE
For webbing characteristics listed as Visual or Visual/AIE the defect shall be clearly visible at normal inspection distance (approximately 3 feet). Certification of acceptance from source is acceptable.				

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.1	TITLE Webbing		SHEET 3 OF 4	DRAWING NUMBER 12961015
				NEXT HIGHER ASSEMBLY
CLASSITIFICAON	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major				
119	Cover fails to coil properly (after releasing).	Level IV	3.2.2	Visual/Manual
120	Edge in lengthwise directions, not curved inward (two places).	Level IV	3.2.2	Visual
121	Dropped knitted stitch on edge (applicable to shuttleless looms only).	Level IV	3.2.1	Visual
122	Catch-cord missing (applicable to shuttleless looms only).	Level IV	3.2.1	Visual
Minor				
201	Single end broken or missing under six (6) inches, but exceeding 1/4 inch.	Level II	3.2.1	Visual/AIE
202	Filling bar coarse or light resulting in visible difference in stiffness or thickness of webbing and extending for 1/4 inch or less in the length direction	Level II	3.2.1	Visual/AIE
203	Drop ply clearly visible on one or two ends within same length and extending over nine linear inches or more.	Level II	3.2.1	Visual/AIE
204	Hitchbhack crack, a clearly visible opening, between adjoining picks, or warpwise tension area over part of the width resulting in visible light and heavy places.	Level II	3.2.1	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.1	TITLE Webbing		SHEET 4 OF 4	DRAWING NUMBER 129761015
				NEXT HIGHER ASSEMBLY 129761016
CLASSITIFICAON	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
205	Three or more floats or skips, less than 1/2 inch in combined warp and filling directions; or single float or skip greater than 1/2 inch but not exceeding one inch, if in warp, or More than ¼ inch of width but not Exceeding one inch, if in filling.	Level II	3.2.1	Visual/AIE
206	More than one knot in any nine linear inches.	Level II	3.2.1	Visual/AIE
207	One knot every two yards with untrimmed ends extending from surface of webbing.	Level II	3.2.1	Visual
208	Single mispick or double pick across the full width.	Level II	3.2.1	Visual
209	Slack end single jerked in between picks or forming clearly visible loops on the surface.	Level II	3.2.1	Visual
210	Slub, or slug, gout more than twice the thickness of the yarn (or ply, if plied).	Level II	3.2.1	Visual/AIE
211	Jerked-in filling, slough-off, slug (a clearly visible loop of filling pulled in at edges).	Level II	3.2.1	Visual
212	Hitchback crack, (clearly visible opening between adjoining picks, or warpwise tension area over part of the width resulting in visible light and heavy places).	Level II	3.2.1	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.2	TITLE Cover, rotating band, flexible (opened)		SHEET 1 OF 1	DRAWING NUMBER 12961016
				NEXT HIGHER ASSEMBLY
CLASSITIFICAON	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Length of band.	Level IV	3.2.2	AIE
102	Width of band.	Level IV	3.2.2	AIE
103	Location of closure patch (two places).	Level IV	3.2.2	AIE
104	Component missing or improperly applied.	Level IV	3.2.2	Visual
105	Closure patch (fastener)	Note 1	3.2.2	certification
106	Distance between the front end of one closure patch (hook) to the rear end of another closure patch (loop)	Level IV	3.2.2	AIE
Minor				
201	Marking missing, misleading or unidentifiable.	Level II	3.2.2	Visual
202	Evidence of poor workmanship.	Level II	3.4	Visual

Note 1: A certificate of material conformance shall be submitted by the contractor for each lot utilized. The submission of the documentation shall be specified in the contract.

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.3	**TITLE** Cover, rotating band, flexible (after closing).		**SHEET** 1 **OF** 1	**DRAWING NUMBER** 12961016 **NEXT HIGHER ASSEMBLY**
CLASSITIFICAON	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical	None defined			
Major				
101	Cover fails to close properly.	Level IV	3.2.3	Visual/Manual
Minor	None defined			

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4.4 Testing.

4.4.1 Stretch of rubber (SPANDEX) yarn covered with spun polyester. Randomly select a roll of spun polyester from the first half of a production shift and a second roll during the last hour of the last half of that production shift per machine. From each sample, accurately measure and cut a ten (10) inch specimen. Stretch it to its maximum length and measure it again. Compare this length to the original length and determine compliance with paragraph 3.3. If any sample fails to meet the applicable requirement, reject the quantities of spun polyester covered rubber represented by the sample.

4.4.2 Assembly of rubber (SPANDEX) yarn covered with spun polyester. Randomly select a roll of spun polyester from the first half of a production shift and a second roll during the last hour of the last half of that production shift per machine. The same samples used in 4.4.1 may be used for this test. From each sample, accurately measure and cut a ten (10) inch specimen. Uncover (remove) the top and bottom polyester cover. Allow the rubber filament to relax (for twenty minutes). Measure the polyester covers (filaments) and the rubber filament for compliance with note 7 drawing 12961015. If any sample fails to meet the applicable requirement, reject the quantities of spun polyester covered rubber represented by the sample.

4.4.3 Full width warp ends of webbing. Determine the full width warp ends of the webbing in accordance with Method 5050 of FED-STD-191.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The flexible rotating band cover (FRBC), covered by this specification, is intended for use in protecting the rotating band and obturator of 155mm projectiles. This specification contains requirements suitable for production.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. For acquisition purposes, the contract or order should specify packaging requirements. The flexible rotating band covers should be palletized and marked in accordance with ASTM-D-3951 (Standard Practice for Commercial Packaging) and be ordered by the applicable stock number.

c. Requirements for submission of first article sample.

d. Submission of contractor acceptance inspection equipment (AIE)(see 6.3).

e. Certificate of material conformance for each lot shipments of product.

f. Suggested sources of FRBC fabricators are:

1. National Industries for the Blind, 1901 North Beauregard St., Suite 200, Alexandria, VA 22311-1727 (Tel (703) 998-0770).

2. Isratex Corp., 523 Kent Ave., Brooklyn, NY 11211 (Tel (718) 599-9100)

3. Star Knitting Corp., 266 Union Ave., Laconia, NJ 03246 (Tel (603) 528-7827)

4. Daun-Ray Corp., P.O. Box 230, 1811 Old E. Stillspring Rd., Tullahoma, TN 37388 (Tel (615) 455-8033).

5. Sherfield Industries, 84 Union Ave., Laconia, NJ 03246 (Tel (603) 524-1401)

6. Creative Apparel Assoc., P.O. Box 208, Belfast, ME 04915 (Tel (207) 342-5830)

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6.3 Submission of contractor acceptance inspection equipment (AIE) designs for approval. Submit copies of designs as required to: Commander, U.S. Army TACOM-ARDEC, ATTN: AMSRD-AAR-QEM-A, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Visual standards. Visual standards for evaluation of defects will be established for each contract after the first month of production.

6.5 Inspection lots. It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria per MIL-STD-1168 and sampling provisions of this specification are maintained.

6.6 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination the characteristic may be measured or gaged to determine acceptability.

6.7 Alternative testing procedures. Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisions specified herein. Prior to applying the alternative procedures, the contractor should describe them in a written proposal and should demonstrate for the approval of the responsible Government technical agency that their effectiveness is equal to, or better than the quality assurance provisions specified herein.

6.8 Textile terminology. The definitions for the textile terms used in this specification may be found in the "Man-made Fiber and Textile Dictionary" issued by the Hoechst Celanese Corporation. Copies of this document may be obtained from Hoechst Celanese Corporation, PO Box 32414, Charlotte, NC 28232-6085

6.9 Definitions

a. Acceptance Inspection Equipment (AIE). Any measuring device that is traceable to the national or international standard which is used to assure conformance of material to the contract requirements.

6.10 Superseding document. QAT-1451 specification is an ARDEC unique specification that is not in ASSIST. If a copy is required it may be requested from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000

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6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes

6.12 Subject term (keyword) listing.

Obturator
Packing
Polyester webbing
Stretch
Unitization
Water Repellency

Custodian:
Army – AR

Preparing Activity:
Army – AR
(Project 1398-2007-001)

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